Graphic Era Hill University

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GEHU/04M/9.1.3

Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no..............................

Name of the Course: BCA/ BCA (AI & DS)
Semester: II
Name of the Paper: Introduction to Operating Systems
Paper Code: TBC 203/ TBD 212
Time: 1.5 hour **Maximum Marks: 50**

Note:
(i) Answer all the questions by choosing any one of the sub-questions
(ii) Each question carries 10 marks.

Q1. (10 Marks)

a. Explain the concept of an Operating System and describe its different types with suitable examples. (CO1)

OR

b. Explain the concept of CPU scheduling and differentiate the working of FCFS and Round Robin CPU scheduling algorithms. (CO2)

Q2. (10 Marks)

a. Explain the different operating system structures such as Monolithic, Layered, Microkernel, and Modular structure. Analyze their advantages and limitations. (CO1)

OR

b. Consider a system which is using the Shortest Job First (SJF) CPU scheduling algorithm to schedule the following set of processes.

Process	CPU-burst (ms)	Arrival time
P1	20	0
P2	25	15
P3	10	30
P4	15	45

Prepare a Gantt chart and calculate CPU utilization, Throughput, Average Turnaround Time, and Average Waiting Time. (CO2)

Q3. (10 Marks)

a. Explain the various services provided by an operating system and discuss how these services help users and application programs. (CO1)

OR

b. Explain Real-Time Operating System (RTOS) with its types. Also describe its characteristics and applications. (CO2)

Q4. (10 Marks)

a. What is a system call? Explain the different types of system calls and illustrate how a system call is used to access operating system services. (CO1)

OR

b. Explain the concept of a process in an operating system and describe the process state

With suitable diagram. (CO2)

Q5. (10 Marks)

a. Differentiate between Microkernel and Monolithic Kernel in tabular form with suitable diagram. (CO1)

OR

b. Consider the following set of processes, and schedule them using the Multilevel feedback queue CPU scheduling algorithm as used in **UNIX SVR3**.

Process	**CPU Burst**	**Arrival time**	**Base Priority**
P1	2	0	60
P2	1	1	60
P3	3	2	60

(CO2)